Ex 14.6 $dz\,dy\,dx = r\,dz\,dr\,d\theta$

Triple Integral in Spherical and cylinderical co-ordinates

1- Evaluate

$$\int_0^{2\pi}\int_0^1\int_0^{\sqrt{1-r^2}} z r \, dz\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^1 \left(\frac{z^2}{2}\right)_0^{\sqrt{1-r^2}} r\, dr\, d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^1 \left[\left(\sqrt{1-r^2}\right)^2 - 0\right] r\, dr\, d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^1 (r - r^3)\, dr\, d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} \left(\frac{r^2}{2} - \frac{r^4}{4}\right)_0^1 d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} \left(\frac{1}{2} - \frac{1}{4}\right) d\theta$$

$$= \frac{1}{8}(\theta)_0^{2\pi} = \frac{1}{\cancel{8}\,4}(\cancel{2}\pi) = \frac{\pi}{4}$$

2- Ex # 1 use triple integration in cylinderical coordinate to find the volume of solid G that is bounded by the hemisphere $z = \sqrt{25 - x^2 - y^2}$ below by the xy-plane, laterally by the cylinderical $x^2 + y^2 = 9$

Sol:-

Cartesian to cylinderical co-ordinates

$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$x^2 + y^2 = r^2$

$dz\,dy\,dx = r\,dz\,dr\,d\theta = dV$

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$$V = \iiint dV$$

$\therefore r = 3$

$\therefore r : 0 \to 3$

$\therefore \theta : 0 \to 2\pi$

$$V = \int_0^{2\pi}\int_0^3\int_0^{\sqrt{25-r^2}} r \, dz \, dr \, d\theta$$

$$V = \int_0^{2\pi}\int_0^3 r \left[z\right]_0^{\sqrt{25-r^2}} dr \, d\theta$$

$$V = \int_0^{2\pi}\int_0^3 r \left[\sqrt{25-r^2}\right] dr \, d\theta$$

$$V = \int_0^{2\pi}\int_0^3 \frac{-1}{2}(25-r^2)^{\frac{1}{2}}(-2r) \, dr \, d\theta$$

$$V = \frac{-1}{2}\int_0^{2\pi} \left[\frac{(25-r^2)^{\frac{3}{2}}}{3/2}\right]_0^3 d\theta$$

$$V = \frac{-1}{3}\int_0^{2\pi} \left[\begin{matrix}(25-9)^{3/2} \\ (25)^{3/2}\end{matrix}\right] d\theta$$

Imp points
upper and lower
dono limits
put krne hain.

$$V = \frac{-1}{3}\int_0^{2\pi} (4^2)^{3/2} - (5^2)^{3/2} \, d\theta$$

$$V = \frac{-1}{3}\int_0^{2\pi} -61 \, d\theta$$

$$V = \frac{61}{3}\left[\theta\right]_0^{2\pi}$$

$$V = \frac{61}{3}\left[2\pi\right] = \frac{122\pi}{3}$$

EX #2
Use cylinderical coordinates to evaluate the integral.

$$\int_3^{-3}\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}}\int_0^{9-x^2-y^2} x^2 \, dz \, dy \, dx$$

Sol:-

$z : 0 \to 9 - x^2 - y^2$

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$z: 0 \to 9 - r^2$

$y = \sqrt{9 - x^2}$

$y^2 = 9 - x^2$

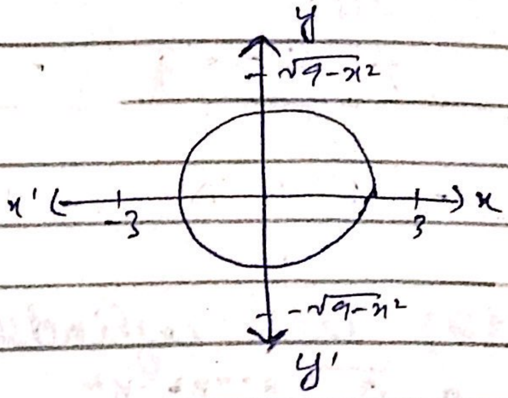
$x^2 + y^2 = 9$

$r^2 = 9$

$r = \pm 3$

$r: 0 \to 3$

$\theta: 0 \to 2\pi$



$$= \int_0^{2\pi}\int_0^{3}\int_0^{9-r^2} r^2 \cos^2\theta \cdot r \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi}\int_0^{3}\int_0^{9-r^2} r^3 \cos^2\theta \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi}\int_0^{3} \Big[r^3 \cos^2\theta \; z \Big]_0^{9-r^2} dr \, d\theta$$

$$= \int_0^{2\pi}\int_0^{3} r^3 \cos^2\theta \, (9 - r^2) \, dr \, d\theta$$

$$= \int_0^{2\pi}\int_0^{3} \left[9r^3 \cos^2\theta - r^5 \cos^2\theta \right] dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{9r^4}{4} \cos^2\theta - \frac{r^6}{6} \cos^2\theta \right]_0^3 d\theta$$

$$= \int_0^{2\pi} \left[\frac{729}{4} \cos^2\theta - \frac{243}{6} \cos^2\theta \right] d\theta$$

$$= \frac{729}{4}\int_0^{2\pi} \cos^2\theta - \frac{243}{6}\int_0^{2\pi} \cos^2\theta \, d\theta$$

$$= \frac{729}{4}\int \left[\frac{1 + \cos 2\theta}{2} \right]_0^{2\pi} - \frac{243}{6}\int \left[\frac{1 + \cos 2\theta}{2} \right]_0^{2\pi}$$

$$= \frac{729}{4}\left[\frac{1}{2}\right](\theta)_0^{2\pi} + \left(\frac{\sin 2\theta}{2}\right)_0^{2\pi} - \frac{243}{6}\left(\frac{1}{2}\right)(\theta)_0^{2\pi} + \left(\frac{\sin 2\theta}{2}\right)_0^{2\pi}$$

$$= \frac{729}{8}(2\pi) - \frac{243}{12}(2\pi)$$

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$$= \frac{729}{4}(\pi) - \frac{243}{2}(\pi)$$

$$= \pi\left(\frac{729}{4} - \frac{243}{2}\right)$$

$$= \frac{243}{4}\pi \quad \text{Ans:-}$$

17) use cylindrical coordinates

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{a^2-x^2-y^2} x^2 \, dz \, dy \, dx$$

Sol)

$$a^2 - x^2 - y^2 = a^2 - r^2$$

$$z : 0 \to a^2 - r^2$$

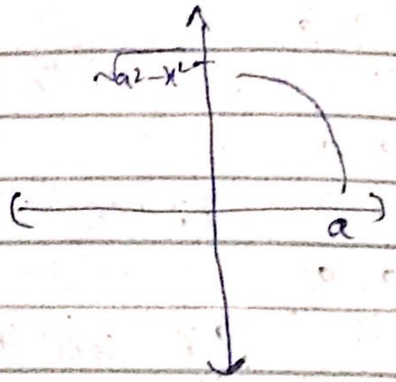
$$y = \sqrt{a^2 - x^2}$$

$$y^2 = a^2 - x^2$$

$$x^2 + y^2 = a^2$$

$$r^2 = a^2$$

$$r = a$$



$$r : 0 \to a$$

$$\theta : 0 \to \frac{\pi}{2}$$

Now,

$$= \int_0^{\frac{\pi}{2}} \int_0^a \int_0^{a^2-r^2} r^2 \cos^2\theta \, r \, dz \, dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^a \int_0^{a^2-r^2} r^3 \cos^2\theta \, dz \, dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^a \left[r^3 \cos^2\theta \, z \right]_0^{a^2-r^2} dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^a \left[r^3 \cos^2\theta \, (a^2 - r^2) \right] dr \, d\theta$$

$$= \int_0^{\pi/2}\int_0^{a} \left[a^2 r^3 \cos^2\theta - r^5 \cos^2\theta\right] dr\, d\theta$$

$$= \int_0^{\pi/2}\int_0^{a} \left[\frac{a^2 r^4 \cos^2\theta}{4} - \frac{r^6 \cos^2\theta}{6}\right]_0^a d\theta$$

$$= \int_0^{\pi/2} \frac{a^6 \cos^2\theta}{4} - \frac{a^6 \cos^2\theta}{6}\, d\theta$$

$$= \frac{a^6}{4}\int_0^{\pi/2} \frac{1+\cos 2\theta}{2}\, d\theta - \frac{a^6}{6}\int_0^{\pi/2} \frac{1\,\text{[redacted]} + \cos 2\theta}{2}\, d\theta$$

$$= \frac{a^6}{4}\left[\left(\frac{1}{2}\right)(\theta)_0^{\pi/2} + \left(\frac{\sin 2\theta}{2}\right)_0^{\pi/2}\right] - \frac{a^6}{6}\left[\left(\frac{1}{2}\right)(\theta)_0^{\pi/2} + \left(\frac{\sin 2\theta}{2}\right)_0^{\pi/2}\right]$$

$$= \frac{a^6}{4}\left[\left(\frac{1}{2}\right)\left(\frac{\pi}{2}\right)\right] - \frac{a^6}{6}\left[\left(\frac{1}{2}\right)\left(\frac{\pi}{2}\right)\right]$$

$$= \frac{a^6}{4}\left[\frac{\pi}{4}\right] - \frac{a^6}{6}\left[\frac{\pi}{4}\right]$$

$$= \frac{a^6\pi}{16} - \frac{a^6\pi}{24}$$

$$= a^6\pi\left(\frac{1}{16} - \frac{1}{24}\right) = \frac{a^6\pi}{48} \quad \text{Ans:-}$$

Q No 2:- Evaluate the iterated integral

$$\int_0^{\pi/2}\int_0^{\cos\theta}\int_0^{r^2} r\sin\theta\, dz\, dr\, d\theta$$

Sol)

$$= \int_0^{\pi/2}\int_0^{\cos\theta} r\sin\theta\, [z]_0^{r^2}\, dr\, d\theta$$

$$= \int_0^{\pi/2}\int_0^{\cos\theta} r\sin\theta\, [r^2]\, dr\, d\theta$$

$$= \int_0^{\pi/2}\int_0^{\cos\theta} r^3 \sin\theta\, dr\, d\theta$$

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$$= \int_0^{\frac{\pi}{2}} \left[\sin\theta \frac{r^4}{4} \right]_0^{\cos\theta} d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{2}} \sin\theta (\cos\theta)^4 \, d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{2}} \cos^4\theta \sin\theta \, d\theta$$

$$= \frac{1}{4} \quad \frac{\cos^5\theta}{5} \longrightarrow \text{Imp}$$

$$= \frac{1}{20} \quad \text{Ans:-}$$

$\frac{1}{4}\frac{1}{5}\left(\cos^5\theta\right)_0^{\frac{\pi}{2}}$

$\frac{1}{20}\left(\cos^5\frac{\pi}{2} - \cos 0^5\right)$

$\boxed{\frac{1}{20}}$

10) The solid that is bounded above by the sphere $x^2+y^2+z^2=1$ and below by the cone $z = \sqrt{x^2+y^2}$.

SoL)

cartesian to cylinderical coordinates

$x = r\cos\theta$

$y = r\sin\theta$

$z = z$

$x^2+y^2 = r^2$

$dz\,dy\,dx = r\,dz\,dr\,d\theta = dv$

As,

$x^2+y^2+z^2 = 1$

$r^2 + z^2 = 1$

$z^2 = 1 - r^2$

$\boxed{z = \sqrt{1-r^2}}$

$z = \sqrt{x^2+y^2}$

$z = \sqrt{r^2}$

$\boxed{z = r}$

$z : r \to \sqrt{1-r^2}$

$r : 0 \to \frac{1}{\sqrt{2}}$

$\theta : 0 \to 2\pi$

$r = 1 - r^2$

$r^2 + r - 1 = 0$

$r^2 + r + \frac{1}{4} - \frac{1}{4} - 1 = 0$

$\left(r + \frac{1}{2}\right)^2 - \frac{5}{4} = 0$

$\left(r + \frac{1}{2}\right)^2 = \frac{5}{4}$

$r + \frac{1}{2} = \frac{\sqrt{5}}{\sqrt{4}}$

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$r = \sqrt{\frac{5}{4}} - \sqrt{\frac{1}{4}}$

$= \sqrt{\frac{5-1}{4}} = \sqrt{\frac{4}{4}}\ 1$

$$= \iiint dV$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\int_r^{\sqrt{1-r^2}} r\, dz\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} \left[r z \right]_r^{\sqrt{1-r^2}} dr d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} r\left[\sqrt{1-r^2} - r\right] dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} \left[r\sqrt{1-r^2} - r^2\right] dr\, d\theta$$

$$= \int_0^{2\pi} \frac{-1}{2}\left[\int_0^{\frac{1}{\sqrt{2}}} -2r\sqrt{1-r^2}\, dr - \int_0^{\frac{1}{\sqrt{2}}} r^2\, dr\right] d\theta$$

$$= \int_0^{2\pi} \frac{-1}{\cancel{2}}\left[\frac{(1-r^2)^{3/2}}{3/\cancel{2}}\right]_0^{\frac{1}{\sqrt{2}}} - \left[\frac{r^3}{3}\right]_0^{1/\sqrt{2}} d\theta$$

$$= \frac{-1}{3}\int_0^{2\pi}\left[(1-r^2)^{3/2}\right]_0^{\frac{1}{\sqrt{2}}} d\theta - \frac{1}{3}\int_0^{2\pi}\left[r^3\right]_0^{1/\sqrt{2}} d\theta$$

$$= \frac{-1}{3}\int_0^{2\pi}\left[\left(1-\frac{1}{\sqrt{2}}\right)^{3/2} - (1)^{3/2}\right] d\theta - \frac{1}{3}\int_0^{2\pi}\left(\frac{1}{\sqrt{2}}\right)^3 d\theta$$

$$= \frac{\pi}{3}(2-\sqrt{2})$$

20) $\displaystyle\int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}}\int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2}\; dz\, dx\, dy$

Sol:- $x = r\cos\theta$

$y = r\sin\theta$

$x^2 + y^2 = r^2$

$dz\,dy\,dx = r\,dz\,dr\,d\theta$

Here:

$x = \sqrt{9-y^2}$

$x = -\sqrt{9-y^2}$

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$y = 3$

$y = -3$

$z = \sqrt{9 - x^2 - y^2}$

$z = -\sqrt{9 - x^2 - y^2}$

So,

$x = \sqrt{9 - y^2}$

Taking square on b.s

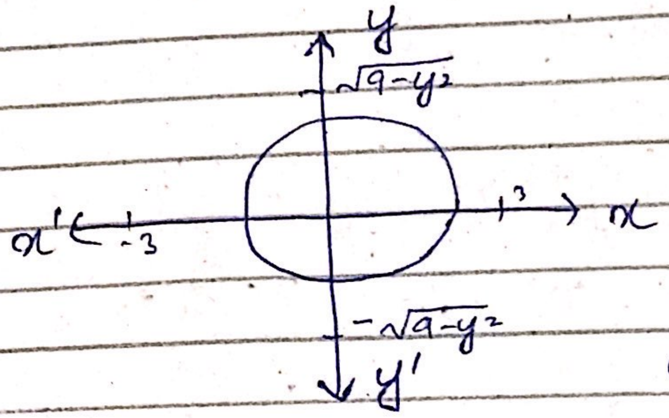
$x^2 = 9 - y^2$

$x^2 + y^2 = 9$

$r^2 = 9$

$r = 3$ $\qquad r : 0 \to 3$

Also,



$\theta : 0 \to 2\pi$

And:-

$z = \sqrt{9 - x^2 - y^2}$

$z = \sqrt{9 - r^2}$

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EX 14.6

Conversion to Spherical Coordinates

$x = \rho \sin\phi \cos\theta$

$y = \rho \sin\phi \sin\theta$

$z = \rho \cos\phi$

$x^2+y^2+z^2 = \rho^2$

ρ = radius of sphere

$\cancel{dz\,dy\,dx = d\rho\, d\phi\, d\theta}$

$dV = dz\,dy\,dx = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

Ex # 4

Use spherical coordinates

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{0}^{\sqrt{4-x^2-y^2}} z^2\sqrt{x^2+y^2+z^2}\,dz\,dy\,dx$$

Sol) $z = \sqrt{4-x^2-y^2}$

$z^2 = 4-x^2-y^2$

$x^2+y^2+z^2 = 4$

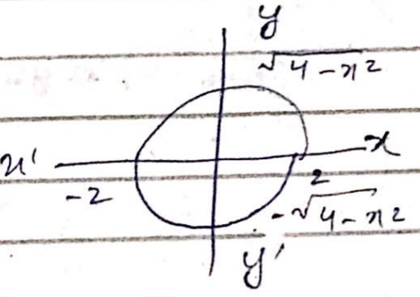
$\rho^2 = 4$

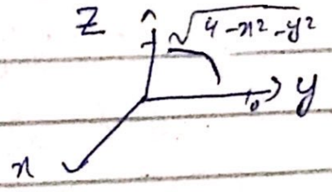
$\rho = 4$

$\rho = 0 \to 2$

$\theta : 0 \to 2\pi$

$\phi : 0 \to \frac{\pi}{2}$





$$= \int_{0}^{2\pi}\int_{0}^{\frac{\pi}{2}}\int_{0}^{2} \rho^2\cos^2\varphi \sqrt{\rho^4}\,(\rho^2 \sin\phi \, d\rho\, d\phi\, d\theta)$$

$$= \int_{0}^{2\pi}\int_{0}^{\frac{\pi}{2}}\int_{0}^{2} \rho^5 \cos^2\phi \sin\phi \, d\rho \, d\phi \, d\theta.$$

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$$= \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \left(\frac{\delta^6}{6}\right)_0^2 \cos^2\phi \sin\phi \, d\phi \, d\theta$$

$$= -\frac{64}{6}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} \cos^2\phi\,(-\sin\phi)\, d\phi \, d\theta$$

$$= -\frac{64}{6}\int_0^{2\pi}\left(\frac{\cos^3\phi}{3}\right)_0^{\frac{\pi}{2}} d\theta$$

$$= -\frac{32}{9}\int_0^{2\pi}\left(\cos^3\frac{\pi}{2} - \cos^3 0\right) d\theta$$

$$= -\frac{32}{9}\int_0^{2\pi}(0-1)\, d\theta$$

$$= \frac{-32}{9}(\theta)_0^{2\pi}$$

$$= \frac{32}{9}(2\pi)$$

$$= \frac{64}{9}\pi$$

20) $$\int_{-3}^{3}\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}}\int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2}\; dz\, dx\, dy$$

Sol)

$z = \sqrt{9-x^2-y^2}$

$z = 9-x^2-y^2$

$x^2+y^2+z^2 = 9$

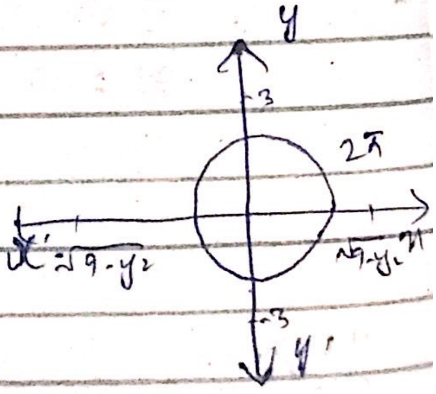
$\delta^2 = 9$

$\delta = 3$

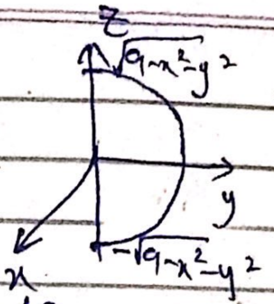
$\delta : 0 \to 3$

$\theta : 0 \to 2\pi$

$\phi : 0 \to \pi$



$\boxed{\begin{matrix}\theta : xy \\ \phi : z\end{matrix}}$ Key point



$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{3} \sqrt{\rho^2}\ \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{3} \rho^3 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\phi \left[\frac{\rho^4}{4}\right]_0^3 d\phi \, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \frac{81\sin\phi}{4} \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \frac{81}{4}\left[-\cos\phi\right]_0^{\pi} d\theta$$

$$= \int_0^{2\pi} \frac{81}{4}\left[-\cos\pi + \cos 0\right] d\theta$$

$$= \int_0^{2\pi} \frac{81}{4}\left[1+1\right] d\theta$$

$$= \frac{162}{4}\int_0^{2\pi} d\theta = \frac{162}{4}\left[\theta\right]_0^{2\pi} = \frac{\overset{81}{\cancel{162}}(\cancel{2}\pi)}{\cancel{4}_{\cancel{2}\,1}}$$

$$\boxed{= 81\pi}$$

18) $\displaystyle\int_{-1}^{1}\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} e^{-(x^2+y^2+z^2)^{3/2}} \, dz \, dy \, dx$

Sol)

$$z = \sqrt{1-x^2-y^2}$$
$$z^2 = 1-x^2-y^2$$
$$x^2+y^2+z^2 = 1$$
$$\rho^2 = 1$$
$$\rho = 1$$
$$\rho : 0 \to 1$$

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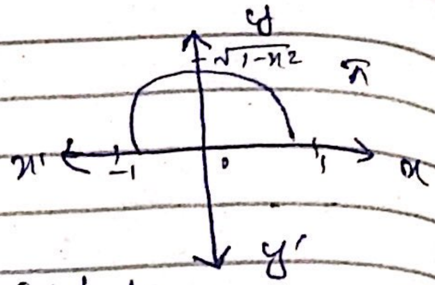
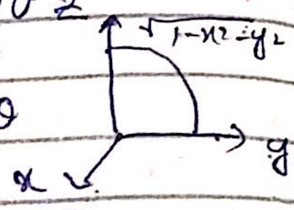
$\theta : 0 \to \pi$

$\phi : 0 \to \frac{\pi}{2}$



$= \int_0^{\pi}\int_0^{\frac{\pi}{2}}\int_0^{1} e^{-(s^2)^{3/2}} \, s^2 \sin\phi \, ds \, d\phi \, d\theta$



$= \frac{-1}{3}\int_0^{\pi}\int_0^{\frac{\pi}{2}}\int_0^{1} e^{-s^3} \, -3s^2 \sin\phi \, ds \, d\phi \, d\theta$

$= \frac{-1}{3}\int_0^{\pi}\int_0^{\frac{\pi}{2}} \left[e^{-s^3}\right]_0^1 \sin\phi \, d\phi \, d\theta$

$= \frac{-1}{3}\int_0^{\pi}\int_0^{\frac{\pi}{2}} \left[e^{-1} - e^{0}\right] \sin\phi \, d\phi \, d\theta$

$= \frac{-1}{3}\int_0^{\pi}\int_0^{\frac{\pi}{2}} \left(\frac{1}{e} - 1\right) \sin\phi \, d\phi \, d\theta$ ← main point

$= \frac{-1}{3}\left(\frac{1-e}{e}\right)\int_0^{\pi} \left[-\cos\phi\right]_0^{\frac{\pi}{2}} d\phi \, d\theta$

$= \frac{-1}{3}\left(\frac{1-e}{e}\right)\int_0^{\pi} \left[-\cos\frac{\pi}{2} + \cos 0\right] d\theta$

$= \frac{-1}{3}\left(\frac{1-e}{e}\right)\int_0^{\pi} 1 \, d\theta$

$= \frac{-1}{3}\left(\frac{1-e}{e}\right)\left[\theta\right]_0^{\pi}$

$= \frac{-1}{3}\pi\left(\frac{1-e}{e}\right)$ Ans:-

$= \left(\frac{e-1}{3e}\right)\pi$ Ans:-

Exercise

Q No 1

Chapter # 15
Topics in Vector Calculus

Gradient:-

$$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} K$$

If ϕ is scalar function

$$\nabla \phi = \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} K$$

Ex #2:-
Find the gradient for $\phi(x, y) = (x+y)$
Sol:-

$$\nabla \phi = \frac{\partial}{\partial x}(x+y) i + \frac{\partial}{\partial y}(x+y) j$$

$$= i + j$$

Divergence of a vector F

$$\vec{F} = f_1 i + F_2 j + F_3 K$$

f_1, f_2, f_3 are 2 or 3 variable for

$$\text{div} F = \nabla \cdot f$$

$$= \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

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curl of vector:-

$\vec{F} = f_1 i + f_2 j + f_3 k$

curl $F = \nabla \times F$

$$= \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_1 & f_2 & f_3 \end{vmatrix}$$

Ex # 4 :- Find the divergence and curl of F

$$F(x,y,z) = \underbrace{x^2 y}_{f_1} i + \underbrace{2y^3 z}_{f_2} j + \underbrace{3z}_{f_3} k$$

$$\text{div } F = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(2y^3 z) + \frac{\partial}{\partial z}(3z)$$

$$= 2xy + 6y^2 z + 3$$

curl of $F = \nabla \times F$

$$= \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 y & 2y^3 z & 3z \end{vmatrix}$$

$$= i(0 - 2y^3) - j(0-0) + k(0 - x^2)$$

$$= -2y^3 i - x^2 k$$

Divergence → scalar only

curl → vector only

Q20) Find the div F and curl F

$F(x,y,z) = e^{xy} i - \cos y j + \sin^2 z K$

Sol)

$\text{Div } F = \nabla \cdot f$

$= \frac{\partial}{\partial x} f_1 + \frac{\partial}{\partial y} f_2 + \frac{\partial}{\partial z} f_3$

$= (e^{xy}) y + \sin y + 2 \sin z \cos z$

$= y e^{xy} + \sin y + 2 \sin z \cos z$

$\text{curl } F = \nabla \times f$

$$= \begin{vmatrix} i & j & K \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^{xy} & -\cos y & \sin z \end{vmatrix}$$

$= i(0-0) - j(0-0) + K(0 - e^{xy}(x))$

$= -x e^{xy} K$

QNO23

Find $\nabla (F \times G)$

$F(x,y,z) = 2xi + j + 4yK$
$G(x,y,z) = xi + yj - zK$

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Sol) $F \times G = \begin{vmatrix} i & j & K \\ 2x & 1 & 4y \\ x & y & -z \end{vmatrix}$

$= i(-z - 4y^2) - j(-2xz - 4xy) + K(2xy - x)$

$\nabla \cdot (F \times G) = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} K\right) \cdot ((-z - 4y^2)i$

$+ j(2xz + 4xy) + K(2xy - x))$

$= \frac{\partial}{\partial x}(-z - 4y^2) + \frac{\partial}{\partial y}(2xz + 4xy)$

$+ \frac{\partial}{\partial z}(2xy - x)$

$= 0 + 4x + 0 \quad = 4x$

QNO 25

Find $\nabla \cdot (\nabla \times F)$

$F(x, y, z) = \sin x \, i + \cos(x - y) j + zK$

Sol) $\nabla \times F = \begin{vmatrix} i & j & K \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \sin x & \cos(x-y) & z \end{vmatrix}$

$= i(0 - 0) - j(0 - 0) + K(-\sin(x - y) - 0)$

$= -\sin(x - y) K$

$\nabla \cdot (\nabla \times F) = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)(-\sin(x-y))$

$= \frac{\partial}{\partial z}(-\sin(x-y))$

$= 0$

conservative field of vector

If $F = \nabla \phi$
then F is called conservative vector field
and ϕ is called the potential function.

Q15) confirm that ϕ is the potential function for given F.

(a) $\phi(x,y) = \tan^{-1}(xy)$

$F(x,y) = \frac{y}{1+x^2y^2} i + \frac{x}{1+x^2y^2} j$

Sol).
If F is conservative than

$\nabla \phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = \frac{y}{1+x^2y^2} i + \frac{x}{1+x^2 y^2} j$

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$\tan^{-1}(xy) = \frac{1}{1+x^2y^2}$

By comparing the coefficients of i and j

$\frac{\partial \phi}{\partial x} = \frac{y}{1+x^2y^2} \rightarrow$ (i)

$\frac{\partial \phi}{\partial y} = \frac{x}{1+x^2y^2} \rightarrow$ (ii)

Integrate eq (1) w.r.t x

$\int \frac{\partial \phi}{\partial x} dx = \int \frac{y}{1+x^2+y^2} dx$

$\phi = \tan^{-1}(xy) + c \rightarrow$ (A)

Integrate eq (2) w.r.t y

$\int \frac{\partial \phi}{\partial y} dy = \int \frac{y}{1+x^2y^2} dy$

$\phi = \tan^{-1}(xy) + c$ (B)

From (A) and (B)

$\phi = \tan^{-1}(xy)$

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Q(16) a)

$\varphi(x, y) = 2y^2 + 3x^2y - xy^3$

$F(x, y) = (6xy - y^3)i + (4y + 3x^2 - 3xy^2)j$

Sol)

If F is conservative

$\nabla\phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k = (6xy - y^3)i + (4y - 3x - 3xy^2)j$

comparing the coefficient of i and j

$\frac{\partial \phi}{\partial x} = (6xy - y^3)i \rightarrow (i)$

$\frac{\partial \phi}{\partial y} = (4y + 3x^2 - 3xy^2)j \rightarrow (ii)$

Integrate (i) w.r.t x

$\int \frac{\partial \phi}{\partial x} = \int (6xy - y^3)\, dx$

$\phi = 3x^2 y - y^3 x$

Integrate (ii) w.r.t y

$$\int \frac{\partial \phi}{\partial y} = \int (4y + 3x^2 - 3xy^2)\, dx$$

$$\phi = 2y^2 + 3x^2y - xy^3$$

From (A) and (B)

$$\boxed{\phi = 3x^2y - xy^3 + 2y^2}$$

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